

Supply Radar — Speaker Notes

Use Case 6 · Supplier & Sourcing Intelligence for Vaighai · a natural, end-to-end talk track you can read from

This document is written the way you would actually **say** it out loud. Read it top to bottom and it walks the whole flow: what the project is, the data, how we clean it, the models and why we chose them, how rain is used, the dashboard (with what to say as you hover each chart), and the behind-the-scenes stack (Postgres, Metabase, n8n). Numbers quoted are from the latest pipeline run.

1. What this is (open with this)

Say it like this: “Vaighai buys coconut-fibre, or coir, from hundreds of small mills across Tamil Nadu. Our sales team always had a clear picture of our customers, but the buying team was flying blind on the supplier side. Supply Radar fixes that — it is basically a radar for our suppliers. For every mill it tells us: is it growing or shrinking, how much of its output are we actually capturing, and where should we be buying more.”

Then give the one-line summary of how: **We pull together three data sources, clean and join them, run a few models to flag risk and forecast volumes, and an AI writes it all up in plain English on a live dashboard.**

2. The data — three sources

Explain that everything rests on three inputs:

- **Purchase records (our ERP):** our own computer records of what we actually bought and received. Exact and reliable.
- **MIR field reports:** our field staff visit mills and estimate, for each one, how much it produced, how much it dispatched to everyone, and how much we took. These are human eyeball estimates — roughly right, not exact.
- **Rainfall:** downloaded free from a weather service (Open-Meteo) for each region.

Say it like this: “The clever part is one simple idea. If we take how much a mill sent out to everyone, and how much of that we took, we get our share of that mill. A big mill where our share is tiny is an opportunity. A mill whose output is falling is a risk. That single ratio is the heart of the whole product.”

3. Cleaning and preparing the data

Walk through what happens before any model runs:

- **Fix the names:** the same mill was written many ways — like ‘A J Coir’ versus ‘AJ Coirs’ — so we match them so they line up.
- **Put everything on the same grain:** one row per mill, per region, per three-month quarter.

- **Remove internal transfers:** about 70% of raw purchase volume is just Vaighai moving its own stock around — that is not the real market, so we drop it.
- **Cap obvious typos:** we cap crazy outlier values, but on a smart log-scale so a genuinely large mill is not flattened — only true fat-finger entries get trimmed.

Say it like this: “In the latest run this turned 208,000 field rows and 13,000 purchase rows into a clean table of about 20,000 mill-quarters covering 1,171 suppliers — and every data-quality gate passed.”

4. The models — names and what each does

There are four models. Two actually learn from data; two are transparent formulas.

Model 1 — Decline Risk (Logistic Regression, calibrated; optional LightGBM challenger)

Predicts the probability a mill will crash next quarter — dispatch falling more than 50% below its own recent average. Output is a percentage and a band: Low, Moderate, or Critical.

Model 2 — Volume Forecast (champion of four methods)

Predicts how many tonnes each mill will dispatch next quarter, plus a low-to-high range. It runs four methods — a linear Ridge model, a seasonal baseline, a blend, and Croston-TSB (built for mills that supply on-and-off) — and keeps whichever was most accurate in testing.

Model 3 — Opportunity Score (a transparent formula)

Ranks where to buy more: mill size, multiplied by how much room is left (headroom), multiplied by recent growth. No black box — anyone can see the logic.

Model 4 — Concentration (HHI + top-5 share)

Tracks how dependent we are on a few suppliers over time — the standard way to measure supplier-dependency risk.

Say it like this: “In the latest run we scored 510 active suppliers, flagged 11 as critical and 242 as moderate risk, and the risk model scored an AUC of about 0.77 on quarters it had never seen.”

5. Why THESE models and not fancier ones

This is the question a reviewer will ask, so meet it head-on. The honest reason is the data is **short and lumpy** — the field data only starts around 2021, so each mill has very few quarters, and many mills supply intermittently with lots of zero quarters.

- **Not Prophet:** it is built for long daily data with rich seasonality. With only a few quarterly points it has nothing to fit and is unreliable.
- **Not ARIMA:** needs a long, stable history per mill; on a handful of quarters it is unstable.
- **Not deep learning / LSTM:** needs thousands of examples, overfits badly here, and is a black box — no good when a buyer asks ‘why was this mill flagged?’
- **Why logistic + ridge + seasonal + Croston:** they are simple, stable on small data, explainable, and Croston is purpose-built for the intermittent mills. We also try LightGBM, but only use it if it genuinely beats the simple model in testing.

Say it like this: “We deliberately kept the models modest, because the history is short and lumpy. A regularized, explainable ensemble with honest cross-validation beats heavy models that would just overfit — and it is something we can actually defend.”

6. How rain is used in the forecast

Coir has to dry, so heavy monsoon rain means wet fibre, drying delays, and mills dispatching less. So each mill-quarter carries the rainfall picture: this quarter’s rain, the previous one or two quarters (because the effect lags), whether it was unusually wet or dry, and — to predict next quarter — a live 16-day forecast plus the normal seasonal rainfall.

Say it like this: “Here is the honest nuance. The predictable part — the monsoon comes every year — is already handled by the seasonal averages. The live 16-day forecast only adds a small near-term correction, and for a full three-month quarter its impact is minor. If we wanted weather to move the forecast more, a season-long monsoon outlook would help more than a 16-day one.”

7. Why it is trustworthy (say this to build credibility)

- **Honest testing:** we use rolling-origin cross-validation — train on the past, test on the next quarter, repeated — so accuracy is measured on the future, not the past it learned from.
- **No leakage:** the seasonal and weather-average clues are computed from training data only, never peeking at the quarter being tested.
- **Calibrated:** when the model says 70% risk, we tune it so that really happens about 70% of the time.
- **Not just accuracy:** we report precision and recall, not only a single score, because a false alarm and a missed collapse feel very different to the buying team.

8. The AI writer (the agent)

After all the numbers are computed, an AI turns them into plain English — this is the part a manager actually reads. It writes a weekly sourcing brief (who to call, who to visit, where to grow) and the on-hover explanations in the dashboard. It only ever sees the final summarized numbers, never raw data, and it is told to use only those numbers and not invent anything. If the AI service is unavailable, a built-in template writes a plainer version so it never breaks.

9. Walking the dashboard — what to SAY as you hover

Open the dashboard and narrate it top to bottom. For each piece, here is the line to say and what to point at.

Top numbers (KPIs)

“These are the headline numbers for the latest quarter — how many suppliers we scored, how many are at critical and moderate risk, the model’s accuracy, and the forecast error.” Point at the 11 critical and the 0.77 accuracy.

The AI brief panel (top card)

“This whole paragraph was written by our AI agent from the numbers — it is the weekly sourcing brief: the headline, who to act on this week, where to grow, and the monsoon outlook.” This is the ‘agent writing in English’ you were looking for.

Top-risk notification (pops top-right)

“On load it alerts us to the mills at critical risk, in plain English, so nothing gets missed.”

Market vs Vaighai chart — HOVER IT

“The top line is the whole market’s output; the lower lines are what we estimated and actually bought. The gap between them is our growth headroom.” As you hover the card, the agent writes a one-line explanation of exactly that.

Decline-risk watchlist — HOVER A ROW

“Every mill with its risk percentage and colour band. Watch — when I hover a row, the agent explains in plain English why this mill is at risk and how much to expect next quarter, with a low-to-high range.” Hover a Critical row like Vishvajith Fiber (was averaging 242 tonnes a quarter, dropped to zero).

Sourcing opportunities — HOVER IT

“These are big mills where our share is tiny — the best places to buy more.” Name the top ones: Vetrivel Pith Block (2,819 tonnes, we take under 3%), Balu Coir (2,501 tonnes, 2%), Palanimurugan Coir (1,720 tonnes, we take zero).

Region mix — HOVER IT

“Where the volume sits by region — mostly Kangeyam, Salem and Pollachi.”

Supply-base concentration — HOVER IT

“This is a good-news story: our reliance on the top-5 suppliers fell from about 91% a few years ago to 51% now — we are far less dependent on a handful of mills.”

Monsoon outlook — HOVER IT

“The next 16 days of rain versus normal per region — heavy rain means wet coir and drying delays.”

10. Behind the scenes — Postgres, Metabase, n8n

Explain this is the optional ‘runs by itself in production’ layer — still fully local, no cloud. Three pieces:

- **PostgreSQL:** a proper database instead of the single local file — the real filing cabinet for production.
- **Metabase:** a point-and-click charting tool on that database — an easy self-serve alternative to the custom dashboard, no coding needed.
- **n8n:** the automation. Think of it as an alarm clock plus a delivery boy — every Monday at 7am it wakes up, reruns the pipeline to retrain and re-score, reads the fresh sourcing brief, and emails it to the purchase managers. No one has to remember to run anything.

Say it like this: “Locally it runs on demand with a simple file database. For production there is a no-cloud stack — Postgres for storage, Metabase for exploring, and n8n that automatically retrains and emails the brief every Monday.”

11. The Azure path (one line for the future)

Close by noting nothing gets rebuilt for the cloud — each piece has a direct Azure twin: n8n becomes Logic Apps, Postgres becomes Azure SQL, the models move to Azure ML, the agent to Azure AI Foundry, Metabase to Power BI, and the email step to Power Automate and Teams.

12. Closing line

Say it like this: “So in one sentence: Supply Radar is a radar for our suppliers — it combines our purchase records, our field team’s market estimates, and rainfall data, then flags which mills are about to shrink, forecasts how much each will supply next quarter, points to where we should buy more, and an AI writes it all up in plain English on a live dashboard — running locally today, and ready to move to Azure.”